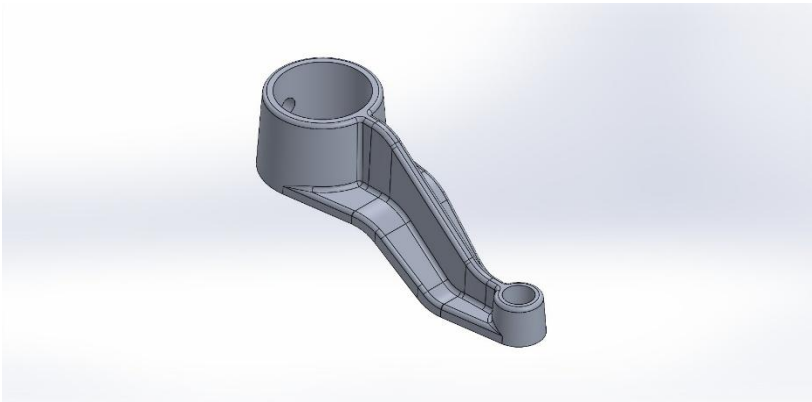




Simulation of Model_Mania_2009

Date: Hidden
Designer: Solidworks
Study name: SimulationXpress Study
Analysis type: Static

Description

Model_Mania_2009-SimulationXpress Study-1

Fixture and Forces were defined as asked for in Phase 2 Instructions.

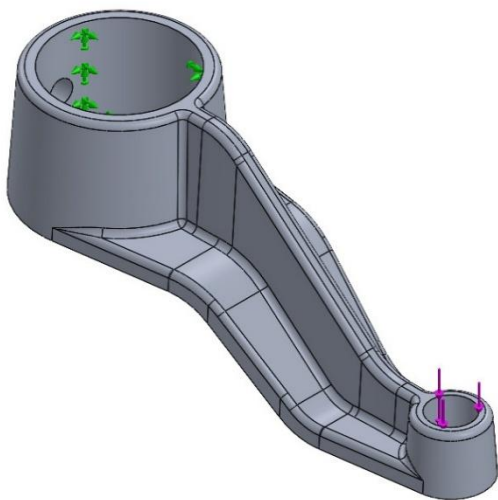
Table of Contents

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Conclusion.....	Error! Bookmark not defined.

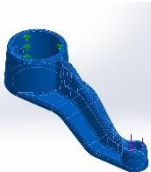


Assumptions

Model Information

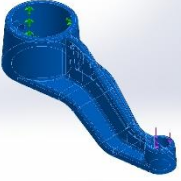


Model name: Model_Mania_2009
Current Configuration: Phase 2

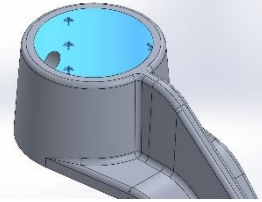
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Fillet1 	Solid Body	Mass:0.228555 kg Volume:8.46501e-05 m^3 Density:2,700 kg/m^3 Weight:2.23984 N	Hidden

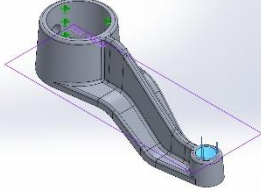


Material Properties

Model Reference	Properties	Components
	Name: 6061-T6 (SS) Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 2.75e+08 N/m ² Tensile strength: 3.1e+08 N/m ²	SolidBody 1(Fillet1)(Model_Mania_2009)

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s), 1 plane(s) Reference: Top Plane Type: Apply force Values: ---, ---, - 1,000 N



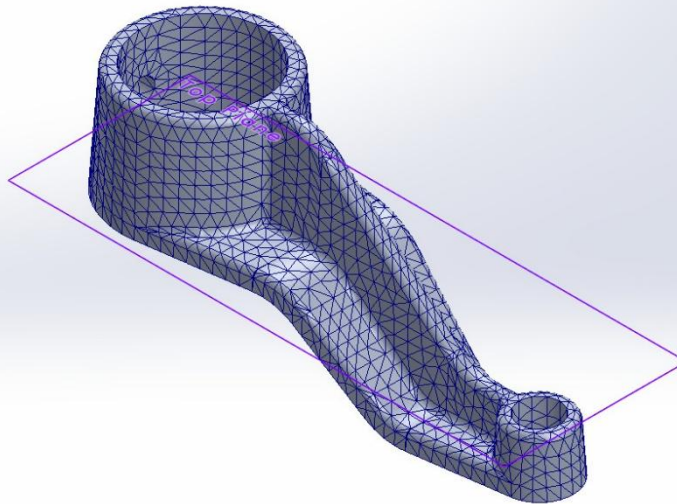
Mesh information

Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	16 Points
Element Size	4.39216 mm
Tolerance	0.219608 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	17063
Total Elements	9891
Maximum Aspect Ratio	24.581
% of elements with Aspect Ratio < 3	96.5
Percentage of elements with Aspect Ratio > 10	0.142
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:03
Computer name:	Hidden

Model name: Model_Mania_2009
Study name: SimulationXpress Study(-Phase 2-)
Mesh type: Solid Mesh



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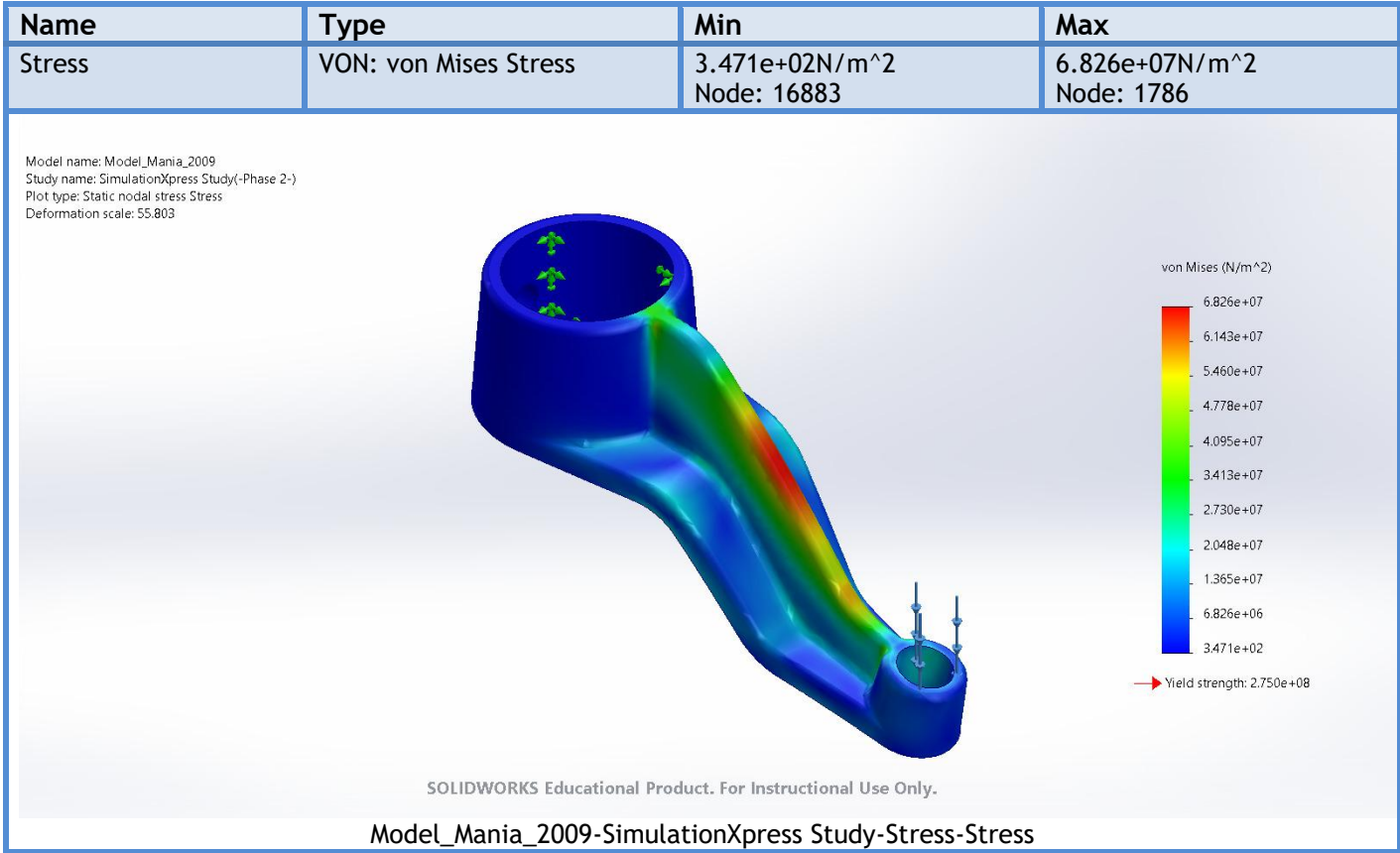
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Model_Mania_2009

4

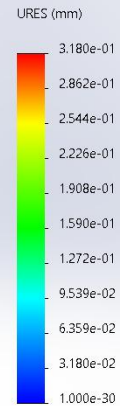
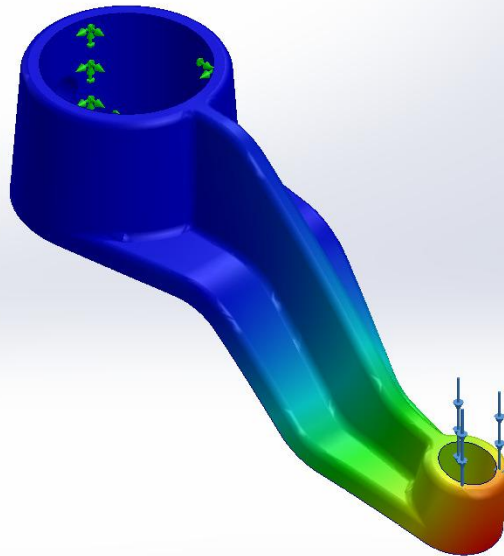
Study Results



Name	Type	Min	Max
Displacement	URES: Resultant Displacement	0.000e+00mm Node: 80	3.180e-01mm Node: 317



Model name: Model_Mania_2009
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Static displacement Displacement
 Deformation scale: 55.803

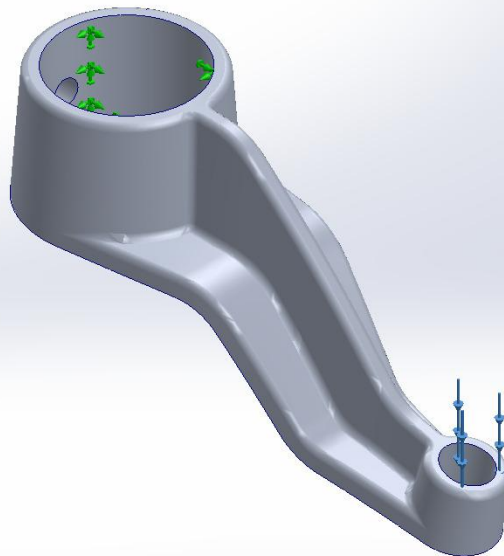


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Model_Mania_2009-SimulationXpress Study-Displacement-Displacement

Name	Type
Deformation	Deformed shape

Model name: Model_Mania_2009
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Deformed shape Deformation
 Deformation scale: 55.803



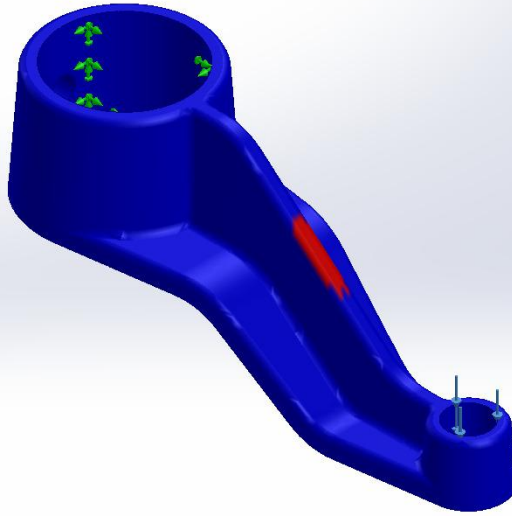
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Model_Mania_2009-SimulationXpress Study-Displacement-Deformation

Name	Type	Min	Max
Factor of Safety	Max von Mises Stress	4.029e+00	7.923e+05



Model name: Model_Mania_2009
Study name: SimulationXpress Study(-Phase 2-)
Plot type: Factor of Safety Factor of Safety
Criterion: Max von Mises Stress
Red < FOS = 4.5 < Blue



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Model_Mania_2009-SimulationXpress Study-Factor of Safety-Factor of Safety

